EXISTENCE OF THE UNIVERSAL FIRST ORDER THICKENING OF A FORMALLY UNRAMIFIED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The leading context span is the author's own prose of the same section that introduces the term universal first order thickening of Z over X, including the solid commutative diagram that displays the defining lifting property. It defines no label and is printed first because the counted statement is a pure existence assertion about that notion.

Editorial note. The author's own remark at source line 1099 that the verification of the cocycle condition is omitted, together with its hint, is retained verbatim; nothing was supplied in its place.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex): a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and the excerpt uses none of those environments; no macro was invented or redefined.

Editorial note. Locators of the cross-chapter pointers refer to a verbatim excerpt of the project's own tag list, sources/stacks-more-morphisms/referenced-tags.txt, each line carrying its stable Stacks tag; the same-chapter pointer refers to its own label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Universal thickening of a ring along a formally unramified ring map. Algebra, Stacks tag 04EB (algebra-lemma-universal-thickening), invoked at proof lines 1036 and 1066. Supplies the affine case: for X = Spec(R) and Z = Spec(S) it produces a first order thickening Z in Z' over X that is universal for affine test diagrams; the passage from affine to arbitrary test diagrams and the glueing of the affine pieces into a global Z' are the retained obligations.
- Glueing of schemes along open subschemes. Schemes, Stacks tag 01JA (schemes-section-glueing-schemes), invoked at proof lines 1098 and 1103. Turns the local thickenings Z_i in Z'_i, together with the transition isomorphisms and their cocycle condition, into a single first order thickening Z in Z' over X.
- Formal unramifiedness is local on source and target. More on Morphisms, Stacks tag 02HC (more-morphisms-lemma-formally-unramified-on-opens), invoked at proof line 1086. Keeps each Z_i to U_i formally unramified for an affine open covering Z = union Z_i with Z_i mapping into an affine

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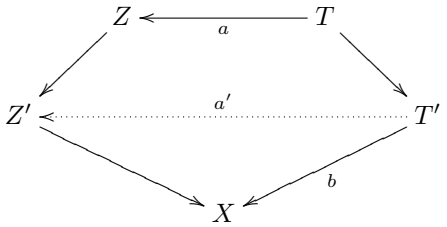
open U_i of X, which is what makes the affine case applicable piece by piece.

Source reference. Original lemma 04EB: sources/stacks-more-morphisms/referenced-tags.txt, source lines 19–19. Affine input cited at proof lines 1036 and 1066; a navigation pointer to the ring-theoretic statement, while the passage from affine to arbitrary test diagrams and the glueing of the pieces stay inside the counted proof.

Source reference. Original lemma 02HC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 856–856. Same-chapter pointer cited at proof line 1086 to keep each Z_i to U_i formally unramified; a localisation statement that the counted proof applies but does not reprove.

Source reference. Original section 01JA: sources/stacks-more-morphisms/referenced-tags.txt, source lines 13–13. Pointer to the glueing formalism invoked at proof lines 1098 and 1103; the transition isomorphisms, the cocycle verification and the descent of the universal property are all in the counted proof.

Let $h : Z \to X$ be a morphism of schemes. A *universal first order thickening* of Z over X is a first order thickening $Z \subset Z'$ over X such that given any first order thickening $T \subset T'$ over X and a solid commutative diagram

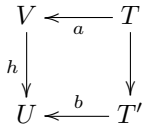


there exists a unique dotted arrow making the diagram commute. Note that in this situation $(a, a') : (T \subset T') \to (Z \subset Z')$ is a morphism of thickenings over X. Thus if a universal first order thickening exists, then it is unique up to unique isomorphism. In general a universal first order thickening does not exist, but if h is formally unramified then it does.

Lemma 0.1. *Let $h : Z \to X$ be a formally unramified morphism of schemes. There exists a universal first order thickening $Z \subset Z'$ of Z over X.*

Proof. During this proof we will say $Z \subset Z'$ is a universal first order thickening of Z over X if it satisfies the condition of the lemma. We will construct the universal first order thickening $Z \subset Z'$ over X by glueing, starting with the affine case which is Algebra, Lemma 04EB. We begin with some general remarks.

If a universal first order thickening of Z over X exists, then it is unique up to unique isomorphism. Moreover, suppose that $V \subset Z$ and $U \subset X$ are open subschemes such that $h(V) \subset U$. Let $Z \subset Z'$ be a universal first order thickening of Z over X. Let $V' \subset Z'$ be the open subscheme such that $V = Z \cap V'$. Then we claim that $V \subset V'$ is the universal first order thickening of V over U. Namely, suppose given any diagram



where $T \subset T'$ is a first order thickening over U. By the universal property of Z' we obtain $(a, a') : (T \subset T') \to (Z \subset Z')$. But since we have equality $|T| = |T'|$ of underlying topological spaces we see that $a'(T') \subset V'$. Hence we may think of (a, a')

as a morphism of thickenings $(a, a') : (T \subset T') \to (V \subset V')$ over U. Uniqueness is clear also. In a completely similar manner one proves that if $h(Z) \subset U$ and $Z \subset Z'$ is a universal first order thickening over U, then $Z \subset Z'$ is a universal first order thickening over X.

Before we glue affine pieces let us show that the lemma holds if Z and X are affine. Say $X = \mathrm{Spec}(R)$ and $Z = \mathrm{Spec}(S)$. By Algebra, Lemma 04EB there exists a first order thickening $Z \subset Z'$ over X which has the universal property of the lemma for diagrams

$$\begin{array}{ccc} Z & \xleftarrow{a} & T \\ {\scriptstyle h}\downarrow & & \downarrow \\ X & \xleftarrow{b} & T' \end{array}$$

where T, T' are affine. Given a general diagram we can choose an affine open covering $T' = \bigcup T'_i$ and we obtain morphisms $a'_i : T'_i \to Z'$ over X such that $a'_i|_{T_i} = a|_{T_i}$. By uniqueness we see that a'_i and a'_j agree on any affine open of $T'_i \cap T'_j$. Hence the morphisms a'_i glue to a global morphism $a' : T' \to Z'$ over X as desired. Thus the lemma holds if X and Z are affine.

Choose an affine open covering $Z = \bigcup Z_i$ such that each Z_i maps into an affine open U_i of X. By Lemma 02HC the morphisms $Z_i \to U_i$ are formally unramified. Hence by the affine case we obtain universal first order thickenings $Z_i \subset Z'_i$ over U_i. By the general remarks above $Z_i \subset Z'_i$ is also a universal first order thickening of Z_i over X. Let $Z'_{i,j} \subset Z'_i$ be the open subscheme such that $Z_i \cap Z_j = Z'_{i,j} \cap Z_i$. By the general remarks we see that both $Z'_{i,j}$ and $Z'_{j,i}$ are universal first order thickenings of $Z_i \cap Z_j$ over X. Thus, by the first of our general remarks, we see that there is a canonical isomorphism $\varphi_{ij} : Z'_{i,j} \to Z'_{j,i}$ inducing the identity on $Z_i \cap Z_j$. We claim that these morphisms satisfy the cocycle condition of Schemes, Section 01JA. (Verification omitted. Hint: Use that $Z'_{i,j} \cap Z'_{i,k}$ is the universal first order thickening of $Z_i \cap Z_j \cap Z_k$ which determines it up to unique isomorphism by what was said above.) Hence we can use the results of Schemes, Section 01JA to get a first order thickening $Z \subset Z'$ over X which the property that the open subscheme $Z'_i \subset Z'$ with $Z_i = Z'_i \cap Z$ is a universal first order thickening of Z_i over X.

It turns out that this implies formally that Z' is a universal first order thickening of Z over X. Namely, we have the universal property for any diagram

$$\begin{array}{ccc} Z & \xleftarrow{a} & T \\ {\scriptstyle h}\downarrow & & \downarrow \\ X & \xleftarrow{b} & T' \end{array}$$

where $a(T)$ is contained in some Z_i. Given a general diagram we can choose an open covering $T' = \bigcup T'_i$ such that $a(T_i) \subset Z_i$. We obtain morphisms $a'_i : T'_i \to Z'$ over X such that $a'_i|_{T_i} = a|_{T_i}$. We see that a'_i and a'_j necessarily agree on $T'_i \cap T'_j$ since both $a'_i|_{T'_i \cap T'_j}$ and $a'_j|_{T'_i \cap T'_j}$ are solutions of the problem of mapping into the universal first order thickening $Z'_i \cap Z'_j$ of $Z_i \cap Z_j$ over X. Hence the morphisms a'_i glue to a global morphism $a' : T' \to Z'$ over X as desired. This finishes the proof. □